Mohadeseh Khoshnevisan

University of Colorado Boulder
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PhD Student, Computer Science

Education

University of Colorado Boulder Spring 2025 – Present
PhD in Computer Science
Current graduate GPA: 3.875/4.00
Supervisor: Prof. Jed Brown

Iran University of Science and Technology (IUST) 2018 – 2022
B.Sc. in Mathematics and Applications, Tehran, Iran
CGPA: 17.24/20 (3.63/4 US GPA); Last 2 Years: 18.45/20 (3.96/4 US GPA)
Thesis: Modeling of ECG Signals Using Spline Interpolation Methods

Honors and Awards

- Graduate Fellowship, University of Colorado Boulder (fully funded PhD with tuition and stipend).
- Exempted from graduate entrance examination and tuition at IUST.
- Selected member of IUST team, National Student Scientific Olympiad.
- Ranked 3rd in cumulative GPA (Top 5%) among 2018 Mathematics entrants at IUST.
- Selected as a Superior Academic Student among top students at IUST.
- Ranked within top 1.5% among 140,000 participants in the National Universities Entrance Exam.
- Admission to NODET (Exceptional Talents) high school.

Research Interests

- Optimization
- Scientific Machine Learning
- Numerical PDEs
- Scientific Computing
- High-Performance Computing (HPC)

Graduate Research

Research Assistant, University of Colorado Boulder 2025 – Present
Supervisor: Prof. Jed Brown

- Working on numerical solvers within the open-source **RDycore** project (performance-portable shallow-water solver).
- Engaging in code development, debugging, and design discussions within weekly lab meetings.

Conference Presentations

Comparison of Roe, HLL, and HLLC Riemann Solvers in RDycore: Accuracy, Diffusion, and GPU Considerations for Shallow-Water Simulations
41st Colorado Conference on Iterative and Multigrid Methods, University of Colorado Boulder 2026

- Presented research on the implementation and evaluation of Roe, HLL, and HLLC Riemann solvers for shallow-water simulations in RDycore.
- Compared solver accuracy, numerical diffusion, and computational performance across representative flooding benchmarks.

Selected CS Projects

- **Efficient Hybrid 3D–2D CNNs for Hyperspectral Image Classification (Python/PyTorch):** Designed hybrid CNN architectures combining 3D and 2D convolutions; evaluated 2D, hybrid, and 3D models under multiple data regimes; demonstrated improved performance–efficiency trade-off of hybrid models.
- **Restaurant3D (C++/OpenGL):** Interactive 3D scene with texture mapping, lighting, and camera controls.
- **Heart Disease Prediction (Python/Scikit-learn):** Built ML models (Logistic Regression, SVM, KNN, Naive Bayes, MLP), with preprocessing and evaluation.
- **Numerical PDE Solvers (Julia):** Chebyshev spectral methods, nonlinear boundary-value solvers, and oscillatory-coefficient diffusion models.

Leadership and Volunteer Experience

- Treasurer, Iranian Students Organization, CU Boulder (2026 – Present).
- Graduate Mentor, Directed Reading Program (DRP), CU Boulder (Fall 2026).
- Graduate Peer Mentor, CU Boulder Graduate School (2026 – Present).
- Core Volunteer, CSEN PhD Open House, CU Boulder (Spring 2026).
- Member, **UPScale Technical Reading Group**, CU Boulder (Fall 2025 – Present).
- Volunteer Mentor for Prospective PhD Applicants, CU Boulder (Fall 2025).
- Mathematics Instructor, NODET High School (2022 – 2024).
- Deputy Secretary, Scientific Society of Mathematics Faculty, IUST (2019 – 2022).
- Financial Advisor & Assistant, Bahar Charity Club, IUST (2020 – 2022).
- Content Producer & Writer, IUST Book Club (2019 – 2020).

Undergraduate Research

- Modeling of ECG Signals Using Spline Interpolation (Python, 2022).
- Differential-equation solvers in Python: Euler, Runge–Kutta, Adams methods, SOR (2022).
- Analysis of geometric transformations and numerical integral methods (2021).
- Black–Scholes PDE modeling and transformation to the heat equation (2021).
- Classification methods using Weka: Decision Tree, Naive Bayes, KNN (2021).
- Fuzzy multi-objective optimization for fire station location (2020).
- Spline interpolation and spline integral methods for air-pollution estimation (MATLAB, 2020).
- Numerical methods in Mathematica: Runge–Kutta, Romberg integration, Legendre/Chebyshev polynomials (2020).

Teaching Experience

Teaching Assistant, University of Colorado Boulder Fall 2026
CSCI 5229 – Computer Graphics

Teaching Assistant, Iran University of Science and Technology (IUST) 2020 – 2022

- Math Lab (Spring 2022)
- Differential Equations (Spring 2022)
- Linear Optimization (Fall 2021)
- Numerical Calculations (Spring 2021)
- General Mathematics I & II (2020 – 2021)

Skills

Programming: Python, C++, C, Julia, MATLAB, Mathematica
Machine Learning: PyTorch, CNNs, model training, evaluation (Accuracy, F1, AA), GPU
Tools: Git, Linux/WSL, VS Code

Scientific Computing: Numerical PDEs, Riemann solvers, spectral methods, finite elements
Languages: Persian (Native), English (Fluent), French (A2)

Selected Mathematics Coursework

- Partial Differential Equations (20/20); Linear Optimization (20/20); Geometry (20/20); Dynamic Systems (20/20); General Topology (19.5/20); Numerical Analysis (18.75/20); Differential Equations (18.5/20); Mathematical Analysis (18/20).

References

Prof. Jed Brown
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